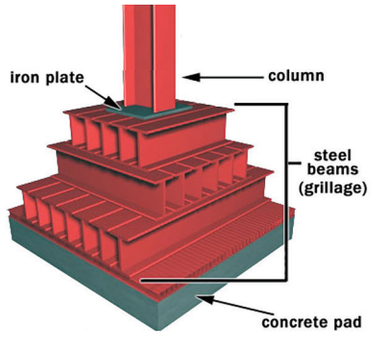
having low bearing capacity. Deep excavation of soil can be avoided by using this foundation. This type of foundation is lighter and more economical.

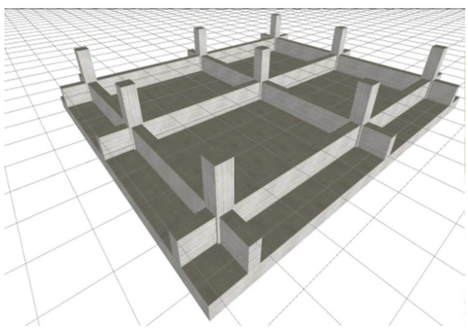
This type of foundation is often used at the base of the steel column. It consists of one, two or more tiers of steel beams superimposed on a layer of concrete. Adjacent tires being placed that right angle to each other.



6.1.4.8. Raft Foundation

A raft or mat is a combined footing that covers the entire area beneath the structure and supports all the columns. The raft foundation is economical than isolated footing, if the sum of the base areas of the isolated footing exceeds about half the total building area.





6.1.5 Deep Foundation

These foundations are suitable when buildings are to be constructed in poor soil (low bearing capacity of soil) or hard rock is only available in high depth. It is classified into two types.

1) Pile foundation
2) Well foundation

6.1.6. Setting Out Of Foundation

It is a process of marking the centre line of a building in the proposed site of construction for earth work excavation.

The following steps are to be followed for setting out of foundation:

i. The land where the building is to be constructed should be cleaned and levelled by removing all the vegetation and undulations.
ii. Dimension of the room to be constructed is studied carefully. Say 4.80 m x 3.30 m.
iii. Now the centreline sketch should be prepared.
iv. If the wall thickness is 0.20 m, the centre-line dimension will be 5.00 m x 3.50m.
v. As per the centre line drawing the centreline of front wall should be marked (points 1 and 2) by driving steel pegs and strings are to be tied to it.
vi. The length of front wall is marked as A and B in that line according to the drawing.
vii. By using right angle, the line is extended from the point B and C is marked as per measurement.
viii. The same procedure is continued until it reaches the point A.
ix. Now, we can get a rectangle bounded by strings.
x. Then the diagonals AC and BD are checked. It should be equal.

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